

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech. (EC) Examination

January- 2014

EC – 701 Advanced High Speed Networks

Date: 04.01.2014, Saturday Time: 10.00 a.m to 1.00 p.m Maximum Marks: 70

Instructions: (1) Attempt all questions.

(2) Answer to the two sections must be written in separate answer books.

(3) Figures to the right indicate full marks

(4) Make suitable assumptions and draw neat figures whenever required.

SECTION –I

- Q-1(a) Explain the operation of TCP/IP reference model with figure. 04
- (b) Give three differences between X.25 and Frame Relay. 03
- Q-2(a) What is the limitation of Bus Topology LAN? Discuss Fiber Channel Protocol Architecture in Brief. 07
- (b) With neat sketch discuss the basic architecture of ATM for an interface between user and network. 07

OR

- Q-2(a) Give the characteristics of high speed gigabit Ethernet and wireless LANs. Explain 10-Gbps Ethernet in detail. 07
- (b) What is emergence of high speed LANs? Give the difference between datagram circuit switching and virtual circuit approach for packet switching techniques. 07
- Q-3 (a) What do you mean by Cryptography? Explain Encryption model for a symmetric key cipher. 07
- (b) Which are differences between Level 2 switch and Hub? Explain the architecture of frame relay. 07

OR

- Q-3(a) Explain the Data Encryption Standard (DES) in detail for security purpose 07
- (b) Briefly describe RSA algorithm. 05
- (c) List the major disadvantages with the layered approach protocol 02

SECTION-II

- Q-4 (a) Differentiate between interior and exterior routing protocol 02
(b) Describe count to infinity problem with necessary illustration. Explain the split horizon with poisoned reverse method to overcome the loop in network 05
Q-5(a) Describe Border Gateway Protocol along with message format 06
(b) Explain the retransmission strategy in TCP 06
(c) Describe MPLS Label Format 02

OR

- Q-5 (a) Describe sending and receiving flow control perspectives 07
(b) Explain Multi Protocol Label Switch Operation 07
Q-6(a) Explain any five RSVP goal and characteristics 05
(b) Describe RSVP Operation. 05
(c) Give the differences between IPv4 and IPv6 04

OR

- Q-6(a) Bifurcate interior routing protocol and exterior routing protocol. Illustrate the network scenario which employs both interior and exterior routing protocol. 05
(b) What are the advantages of adaptive routing over fixed routing? Explain fluttering in detail. 05
(c) Explain RTT Variance Estimation 04